Stability-Based Early Stopping for Heterogeneous Effects in Sequential A/B Experiments

Mayuri Chatterjee

Abstract

Sequential A/B experiments are often monitored using population-average treatment effects. When treatment effects are heterogeneous, however, an aggregate effect can conceal clinically or operationally important harm in a subset of the population. Recent work on heterogeneous early stopping addresses this problem by explicitly modeling treatment-effect heterogeneity. A complementary question is whether an estimated affected population is stable enough across interim analyses to support a reproducible decision.

This paper studies a stability-augmented stopping rule for heterogeneous treatment effects. At each interim analysis, a logistic treatment-by-covariate model is used to estimate the conditional average treatment effect (CATE) on the probability scale. Individuals with estimated CATE at or below -0.02 are classified as experiencing meaningful harm. The baseline stopping rule is triggered when at least 50% of the currently observed population is classified as harmed. The stability-augmented rule additionally requires the Jaccard similarity between consecutive estimated harmful populations to exceed a prespecified threshold.

The study is a controlled simulation investigation rather than a proposal for a formally calibrated sequential hypothesis test. We simulate 500 replicated experiments of size $N = 4000$, with equal randomization and six cumulative interim analyses at sample sizes 1000, 1600, 2200, 2800, 3400, and 4000. Two heterogeneous data-generating mechanisms are considered: a threshold-based harmful subgroup and a smooth logistic treatment-effect surface.

The stability requirement has its largest effect when the estimated harmful population is unstable. At the illustrative threshold $J_0 = 0.75$, the stability-augmented rule reduces the stopping probability in the threshold scenario from 39.2% to 31.4%, while increasing agreement with the final analysis from 79.2% to 85.4% and reducing decision reversal from 20.8% to 13.8%. In the smooth scenario, where the estimated harmful population is substantially more persistent, the corresponding changes are small. Across stability thresholds, stronger persistence requirements reduce reversal but increase the amount of data required before stopping.

The results support stability as a useful diagnostic dimension for sequential decisions involving heterogeneous effects. They do not establish an optimal Jaccard threshold or provide formal Type-I error control; those questions require a future inferential development beyond the present simulation study.

1 Introduction

Randomized A/B experiments provide a direct framework for estimating the causal effect of an intervention. In a conventional analysis, attention is often placed on the average treatment effect (ATE), which summarizes the population-level difference between treatment and control. The ATE is useful when treatment effects are approximately homogeneous, but it can be misleading when the effect varies systematically across individuals or subgroups.

The potential-outcomes framework makes this distinction explicit [5]. Let $A_i \in \{0, 1\}$ denote treatment assignment and $Y_i(1)$ and $Y_i(0)$ the corresponding potential outcomes. The individual treatment effect is

$$\tau_i = Y_i(1) - Y_i(0),$$

while the conditional average treatment effect is

$$\tau(x) = \mathbb{E}\{Y(1) - Y(0) \mid X = x\}.$$

Thus, treatment effectiveness can depend on baseline characteristics rather than being represented adequately by a single scalar parameter.

A substantial methodological literature has developed around the estimation and evaluation of treatment-effect heterogeneity. Tree-based methods provide interpretable approaches for discovering heterogeneous causal effects [2], while more recent reviews emphasize the distinction between exploratory subgroup discovery, CATE estimation, treatment-regime construction, and rigorous evaluation of data-driven subgroups [3]. False discovery is a particular concern when many candidate subgroup structures are considered; simulation evidence shows that naive subgroup procedures can produce large numbers of spurious HTE findings [4].

Sequential experimentation adds another dimension. An experiment may be examined repeatedly as data accumulate, and an early stopping decision may be made when evidence of harm becomes sufficiently strong. The problem becomes more difficult under treatment-effect heterogeneity because a treatment can be harmful for one population segment while appearing relatively benign on average. Adam et al. [1] formalize this problem and show that conventional homogeneous stopping procedures can have low stopping probability when harm is concentrated in a minority population. Their CLASH framework explicitly targets heterogeneous early stopping. Related work has also developed online testing procedures for heterogeneous treatment effects [6].

The present study focuses on a narrower and complementary question. Suppose an interim analysis identifies a population whose estimated treatment effect is meaningfully harmful. The estimated population itself may change as additional observations arrive. An early stopping decision may therefore depend not only on the magnitude of the estimated effect, but also on the persistence of the population being classified as harmed.

We investigate whether this persistence can be incorporated into a transparent stability criterion. Specifically, we compare a baseline CATE-based stopping rule with a rule that additionally requires the estimated harmful population to remain sufficiently similar across consecutive interim analyses. Similarity is quantified using the Jaccard index.

The study has three objectives:

1. quantify how unstable CATE-based harm classifications can be under different forms of treatment-effect heterogeneity;
2. determine how a stability requirement changes stopping probability, decision reversal, and stopping sample size; and
3. characterize the trade-off between early stopping and persistence of the estimated affected population.

The purpose is explicitly exploratory. The study does not claim that the proposed rule is an optimal sequential test, nor does it establish formal Type-I error control. Instead, it provides a reproducible simulation benchmark for understanding whether population stability contains information that is not captured by the harmful fraction alone.

2 Statistical framework

2.1 Randomized experiment

Consider a randomized experiment with observations

$$(X_i, A_i, Y_i), \qquad i = 1, \ldots, N,$$

where X_i is a baseline covariate, A_i is treatment assignment, and $Y_i \in \{0, 1\}$ is a binary outcome. Treatment is assigned independently according to

$$A_i \sim \text{Bernoulli}(0.5).$$

The population-average treatment effect is

$$\text{ATE} = \mathbb{E}\{Y(1) - Y(0)\}.$$

Although the ATE remains an important summary, the analysis here is concerned with the conditional effect

$$\tau(x) = \mathbb{E}\{Y(1) - Y(0) \mid X = x\}.$$

2.2 CATE estimation

At each interim analysis, the treatment effect is estimated using the interpretable logistic model

$$\text{logit}\{P(Y = 1 \mid A, X)\} = \beta_0 + \beta_A A + \beta_X X + \beta_{AX} AX.$$

For a covariate value x, the fitted probabilities under treatment and control are

$$\widehat{p}_1(x) = P_{\widehat{\beta}}(Y = 1 \mid A = 1, X = x)$$

and

$$\widehat{p}_0(x) = P_{\widehat{\beta}}(Y = 1 \mid A = 0, X = x).$$

The estimated probability-scale CATE is

$$\widehat{\tau}(x) = \widehat{p}_1(x) - \widehat{p}_0(x).$$

This estimator is deliberately simple and transparent. It is not intended to represent the full range of modern HTE estimators. Its role in this study is to provide a common, interpretable estimator across all interim analyses and simulation scenarios.

Approximate CATE standard errors are obtained using the delta method in the associated R implementation. These uncertainty estimates are retained for validation, but the stopping rules studied here are based on point estimates rather than formal confidence-bound or alpha-spending criteria.

3 Harm classification and stopping rules

3.1 Meaningful harm

A treatment effect is classified as meaningfully harmful when the estimated probability-scale CATE satisfies

$$\widehat{\tau}(X_i) \leq -0.02.$$

The threshold of -0.02 corresponds to a two-percentage-point reduction in the outcome probability. It is a simulation parameter and is not intended as a generally applicable clinical or business decision threshold.

At interim analysis k, let

$$\widehat{\mathcal{H}}_k = \{i : \widehat{\tau}_k(X_i) \leq -0.02\}$$

denote the estimated harmful population, and define the harmful fraction as

$$\widehat{h}_k = \frac{|\widehat{\mathcal{H}}_k|}{n_k},$$

where n_k is the cumulative sample size available at interim k.

3.2 Baseline CATE-based stopping rule

The baseline rule stops when the estimated harmful fraction reaches

$$h_0 = 0.50.$$

Thus,

$$\text{Stop}_k^{\text{naive}} = \mathbb{I}\left(\widehat{h}_k \geq h_0\right).$$

This rule intentionally isolates the contribution of the harmful-population threshold. It is not presented as a formally calibrated sequential test.

3.3 Stability of the estimated harmful population

The key extension concerns the composition of the estimated harmful population. Consider consecutive interim estimates $\widehat{\mathcal{H}}_{k-1}$ and $\widehat{\mathcal{H}}_k$. Their Jaccard similarity is

$$J_k = \frac{|\widehat{\mathcal{H}}_{k-1} \cap \widehat{\mathcal{H}}_k|}{|\widehat{\mathcal{H}}_{k-1} \cup \widehat{\mathcal{H}}_k|}.$$

The index satisfies

$$0 \leq J_k \leq 1,$$

with larger values indicating greater overlap.

The stability-augmented rule requires both sufficient harmful fraction and sufficient persistence:

$$\mathrm{Stop}_k^{\mathrm{stable}} = \mathbb{I}\left(\widehat{h}_k \geq h_0 \text{ and } J_k \geq J_0\right).$$

The sensitivity analysis considers

$$J_0 \in \{0.50, 0.60, 0.70, 0.75, 0.80, 0.90\}.$$

The value $J_0 = 0.75$ is used as an illustrative operating point in the main comparison. It is not estimated or optimized from the simulation results.

3.4 Decision reversal

Because no external truth is available during a real sequential experiment, the simulation uses the final interim analysis at $n = 4000$ as a reference point for assessing instability.

A reversal occurs when an early stopping decision based on the harmful classification is not supported by the corresponding final classification. This quantity measures the persistence of the sequential decision rather than formal predictive validity against an external population truth.

4 Simulation design

4.1 Common design

The simulation contains 500 replicated randomized experiments. Each experiment has

$$N = 4000$$

observations and equal treatment allocation.

Cumulative interim analyses occur at

$$n_k \in \{1000, 1600, 2200, 2800, 3400, 4000\},$$

corresponding to information fractions

$$0.25,\ 0.40,\ 0.55,\ 0.70,\ 0.85,\ 1.00.$$

The baseline covariate in Scenario A is generated as

$$X \sim N(0,1).$$

The simulation compares two qualitatively different heterogeneous effect structures. This contrast is intentional: the objective is to determine whether stability-based stopping behaves differently when the underlying harmful population has a sharp boundary versus a smooth transition.

4.2 Scenario A: threshold heterogeneity

The first scenario contains a sharply defined harmful subgroup. Let $q_{0.75}$ denote the upper quartile of the standard normal distribution and define

$$\mathcal{H} = \{X > q_{0.75}\}.$$

Approximately 25% of the population belongs to this subgroup. Under control, the outcome probability is

$$p_0 = 0.10.$$

Treatment has no effect outside the harmful subgroup and reduces the outcome probability by 0.04 within the subgroup:

$$\tau(X) = \begin{cases} 0, & X \leq q_{0.75}, \\ -0.04, & X > q_{0.75}. \end{cases}$$

The resulting population-average treatment effect is approximately

$$\text{ATE} \approx -0.01.$$

This scenario is deliberately challenging for a smooth parametric CATE estimator because the true effect changes discontinuously at the subgroup boundary. The simulation therefore provides a setting in which estimated harmful classifications may move substantially as the sample grows.

4.3 Scenario B: smooth logistic heterogeneity

The second scenario generates treatment-effect heterogeneity from a treatment-by-covariate logistic interaction. The outcome model is

$$\text{logit}\{P(Y = 1 \mid A, X)\} = \beta_0 + \beta_X X + \beta_A A + \beta_{AX} AX,$$

with the same functional form used by the analysis model.

The covariate is generated on a bounded interval, $X \in [-1, 1]$, and the parameters are chosen so that treatment is harmful throughout the simulated range while the magnitude of harm varies smoothly with X.

The resulting true CATE ranges approximately from -0.062 to -0.014, with an average treatment effect of approximately

$$\text{ATE} \approx -0.037.$$

Unlike Scenario A, there is no sharp subgroup boundary. The estimated harmful population is therefore expected to be more persistent as observations accumulate.

4.4 Simulation implementation

At each interim analysis, only observations available by that point are used to fit the logistic CATE model. Counterfactual predictions under treatment and control are generated for every observed individual, and the CATE is computed from their difference.

The simulation code also retains the known true CATE values. These values are used only for post-simulation validation and are never supplied to the estimation or stopping procedure.

5 Evaluation criteria

The simulation evaluates four operating characteristics.

5.1 Stopping probability

For each scenario and stopping rule, the stopping probability is the proportion of replicated experiments that stop at any interim analysis.

5.2 Agreement with the final analysis

For each replicate, the final interim analysis at $n = 4000$ is treated as a reference classification. Agreement measures the proportion of early stopping decisions that remain consistent with that final analysis.

This is a stability diagnostic rather than a gold-standard measure of correctness.

5.3 Decision reversal

Decision reversal measures the proportion of early stopping decisions that are not supported by the final interim analysis. Lower reversal indicates greater persistence of the sequential decision.

5.4 Stopping sample size

For experiments that stop, the stopping sample size is the first interim sample size at which the corresponding stopping rule is satisfied. The mean stopping sample size summarizes the information cost associated with the stability requirement.

6 Results

6.1 Estimated harmful fraction

Figure 1 summarizes the mean estimated harmful fraction over the six interim analyses.

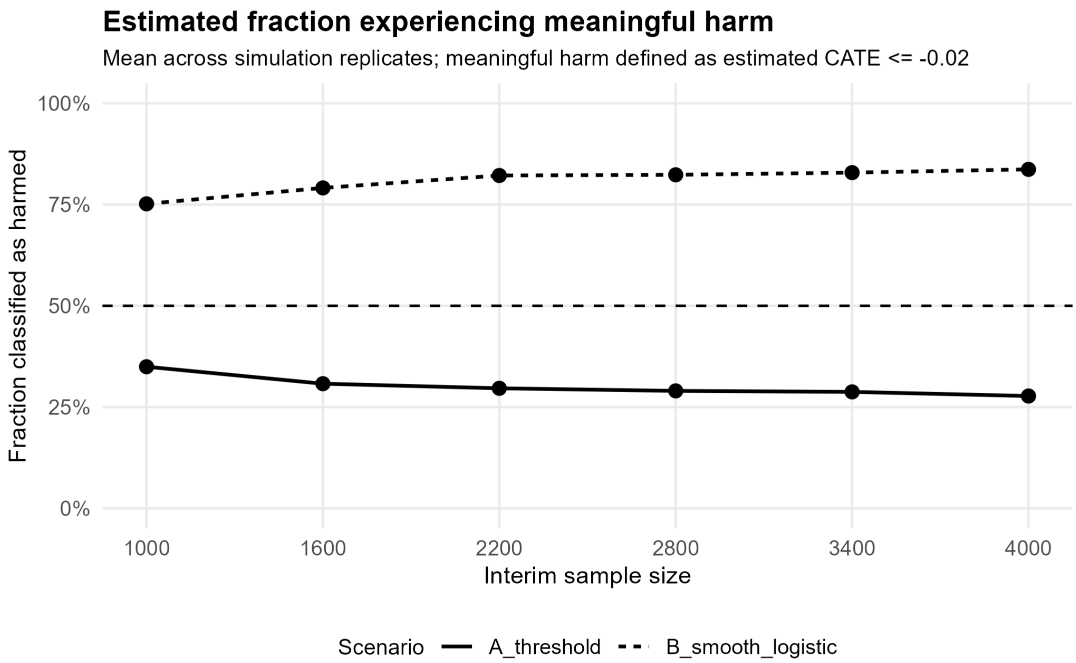

Figure 1: Mean estimated fraction of observations classified as meaningfully harmed across interim analyses. Meaningful harm is defined as $\widehat{\tau}(X) \leq -0.02$; the horizontal reference corresponds to the 50% harmful-fraction threshold.

In Scenario A, the estimated harmful fraction decreases from approximately 35% at the first interim analysis to approximately 28% at the final analysis. In Scenario B, the estimated harmful fraction is substantially larger, increasing from approximately 75% to approximately 84%.

The difference is important because both scenarios can generate early evidence of harm, but the population classified as harmed behaves very differently as the experiment accumulates information.

6.2 Stability of the estimated harmful population

Figure 2 displays the mean Jaccard similarity between consecutive estimated harmful populations.

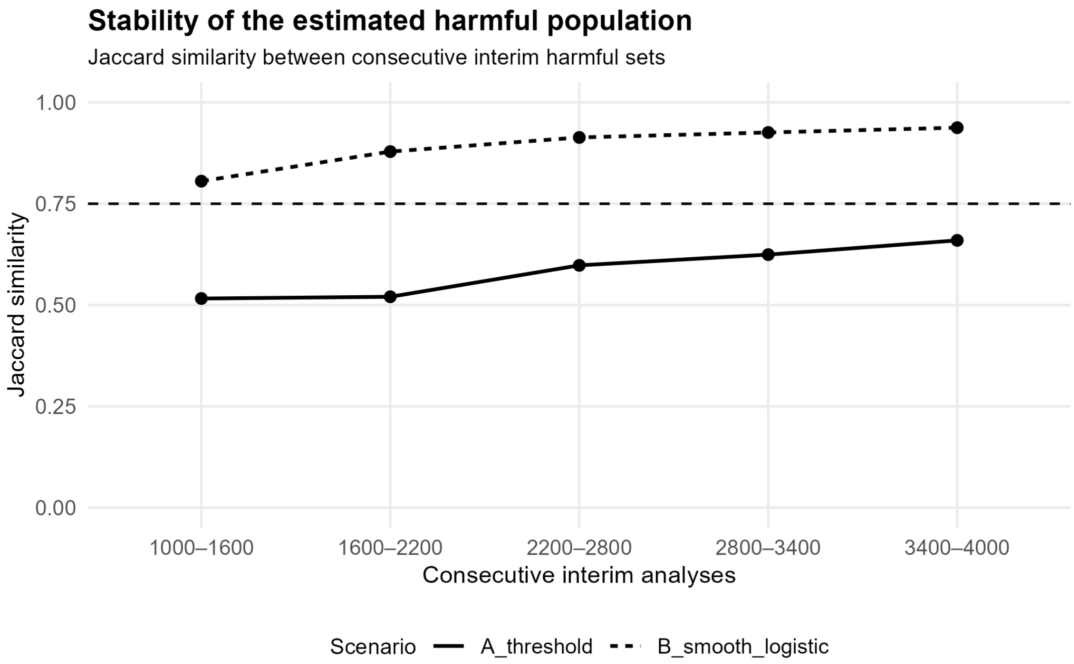

Figure 2: Jaccard similarity between consecutive estimated harmful populations. The horizontal reference indicates the illustrative threshold $J_0 = 0.75$.

Scenario A exhibits substantially lower persistence. The Jaccard similarity starts at approximately 0.52 and increases to approximately 0.66 by the final transition, remaining below the illustrative threshold of 0.75.

Scenario B is considerably more stable. Jaccard similarity increases from approximately 0.81 to approximately 0.94 across the observed transitions.

Thus, the scenarios differ not only in the magnitude of the estimated harmful fraction, but also in the persistence of the population being classified as harmed.

6.3 Sensitivity of stopping probability to the stability threshold

Figure 3 shows the stopping probability under different values of J_0.

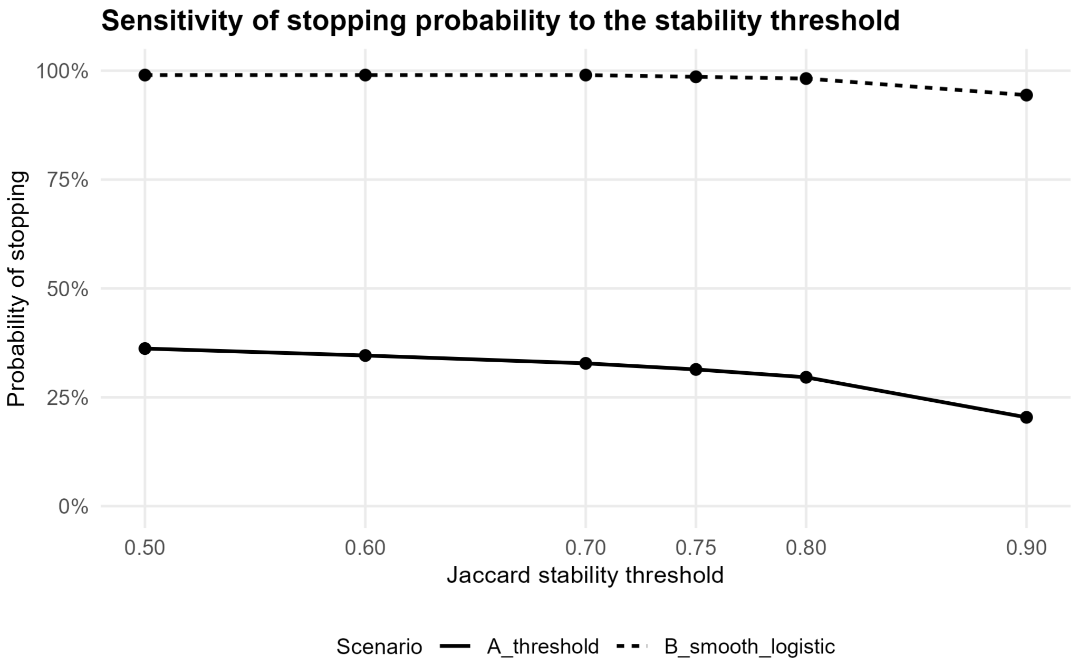

Figure 3: Probability of stopping under the stability-augmented rule as a function of the Jaccard threshold.

In Scenario A, increasing J_0 progressively reduces the probability of stopping, from 36.2% at $J_0 = 0.50$ to 20.4% at $J_0 = 0.90$. In Scenario B, the stopping probability remains 99.0% through $J_0 = 0.70$ and declines to 94.4% at $J_0 = 0.90$.

The effect of the stability requirement is therefore concentrated in the scenario where the estimated harmful population is least persistent.

6.4 Decision reversal

Figure 4 shows the probability of decision reversal.

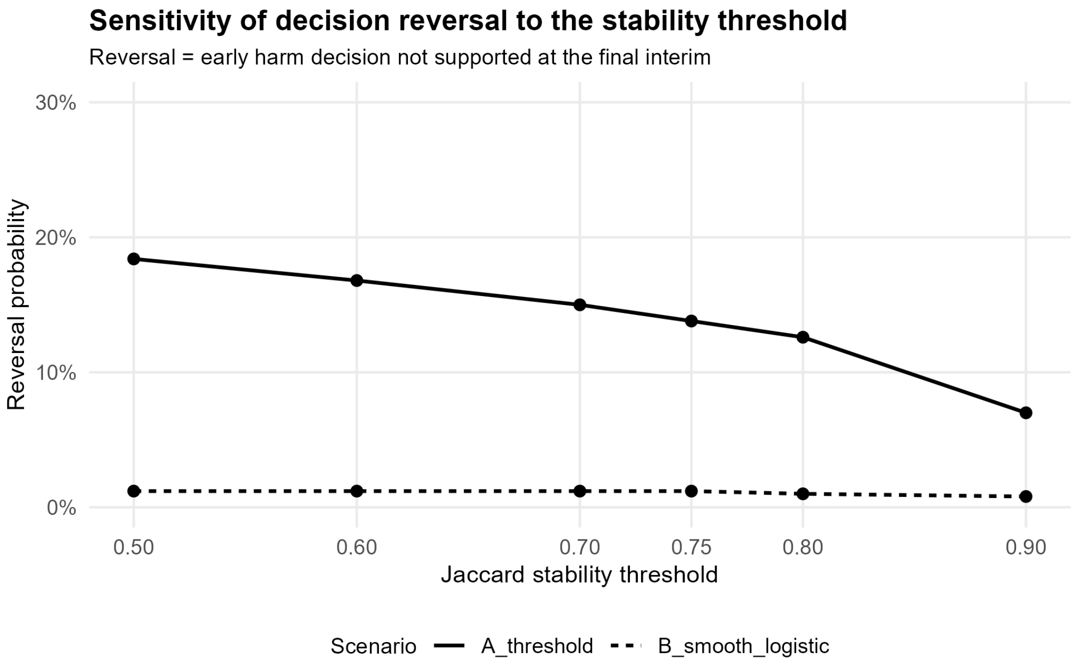

Figure 4: Probability of decision reversal as a function of the Jaccard threshold. A reversal occurs when an early stopping decision is not supported by the final interim analysis.

For Scenario A, reversal decreases from 18.4% at $J_0 = 0.50$ to 7.0% at $J_0 = 0.90$. In Scenario B, reversal remains close to 1% across the range of thresholds.

The result indicates that increasing the persistence requirement can suppress early decisions that are not stable, but that the magnitude of this benefit depends strongly on the underlying treatment-effect structure.

6.5 Stopping-time trade-off

The reduction in reversal is accompanied by a larger information requirement. Figure 5 shows the mean stopping sample size.

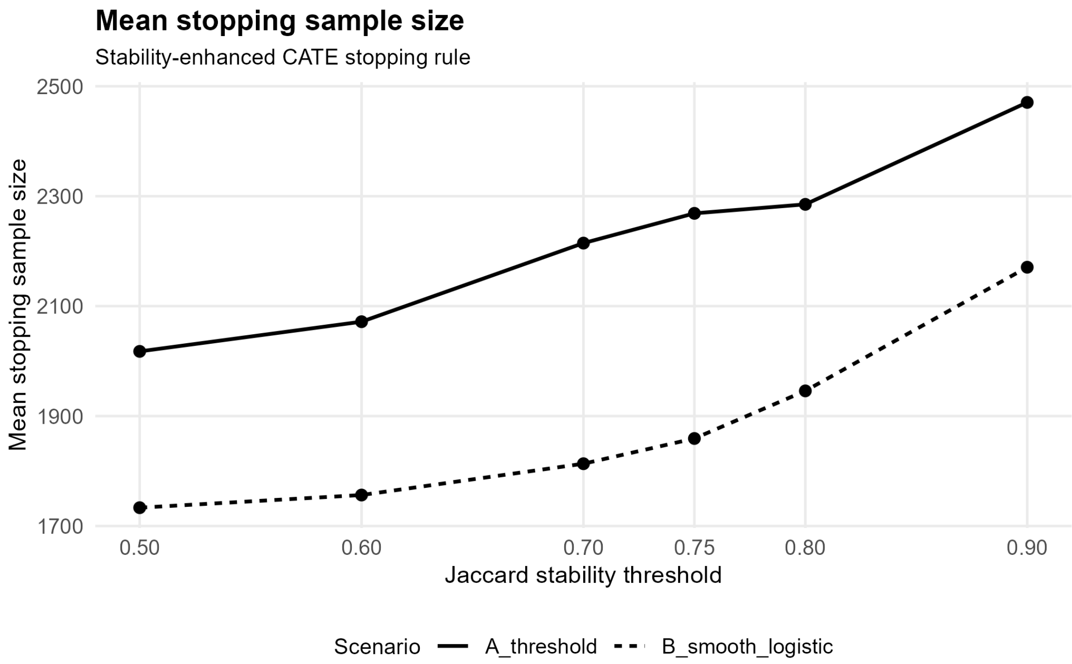

Figure 5: Mean stopping sample size under the stability-augmented rule as a function of the Jaccard threshold.

For Scenario A, the mean stable stopping sample size increases from approximately 2,018 at $J_0 = 0.50$ to approximately 2,471 at $J_0 = 0.90$. For Scenario B, it increases from approximately 1,733 to approximately 2,171.

The stability requirement therefore induces a clear efficiency–stability trade-off.

6.6 Illustrative operating point

For a focused comparison, Table 1 reports the operating characteristics at $J_0 = 0.75$.

Table 1: Operating characteristics at the illustrative stability threshold $J_0 = 0.75$.

	Scenario A	Scenario B
Naive stopping probability	39.2%	99.0%
Stable stopping probability	31.4%	98.6%
Naive agreement with final analysis	79.2%	98.8%
Stable agreement with final analysis	85.4%	98.4%
Naive reversal probability	20.8%	1.2%
Stable reversal probability	13.8%	1.2%
Naive mean stopping sample size	1,898	1,677
Stable mean stopping sample size	2,269	1,859

In Scenario A, adding the stability requirement reduces stopping probability from 39.2% to 31.479.2% to 85.4the mean stopping sample size from 1,898 to 2,269.

In Scenario B, the stability requirement has little effect on the decision process: stopping remains above 98%, and reversal remains approximately 1%. The corresponding increase in mean stopping sample size is smaller in absolute terms than in Scenario A.

7 Discussion

The simulation highlights a distinction between two aspects of heterogeneous early stopping. The first is the magnitude of the estimated harmful population. The second is the persistence of the population being classified as harmful.

This distinction is relevant to the broader literature on heterogeneous early stopping. Adam et al. [1] show that aggregate stopping procedures can have poor stopping probability when harm is concentrated in a subpopulation, motivating methods that explicitly account for heterogeneity. The present simulation does not reproduce or implement CLASH. Instead, it takes the estimated harmful population as given and asks whether the population itself remains stable enough to support an early decision.

The threshold scenario provides the clearest demonstration. The true treatment-effect structure contains a sharp subgroup boundary, whereas the analysis uses a smooth logistic interaction. The resulting model-based classification can therefore move as additional observations are incorporated. The stability requirement filters some of these transient classifications, reducing reversal at the cost of delayed stopping.

The smooth scenario behaves differently. Because the data-generating mechanism is aligned with the analysis model, the estimated treatment-effect surface is more persistent. Consequently, imposing a stability requirement does not substantially alter the stopping decision.

This interpretation is consistent with the broader HTE literature, which emphasizes that subgroup identification and treatment-effect estimation are sensitive to model specification, search complexity, and the way candidate subgroups are evaluated [3, 4]. In the present setting, the Jaccard index is not a substitute for formal post-selection inference or multiplicity control. It is a descriptive measure of whether a data-driven population classification persists over time.

The results also clarify the role of the threshold J_0. There is no universally optimal value in these simulations. A larger threshold produces a more conservative decision rule: it reduces reversal but increases the information required before stopping. The appropriate operating point would therefore depend on the relative costs of premature stopping, delayed stopping, and decision instability in the application.

The simulation should consequently be interpreted as a proof-of-concept for the stability dimension rather than as evidence that $J_0 = 0.75$ is a generally optimal choice.

8 Limitations

Several limitations constrain the interpretation of the results.

First, the simulation uses a single baseline covariate and an interpretable logistic treatment-by-covariate model. Modern HTE analyses may involve many covariates, nonlinearities, machine-learning estimators, missing data, or complex dependence structures. The behavior of the stability criterion under such estimators remains unknown.

Second, the harmfulness threshold of -0.02 and harmful-fraction threshold of 50% are deliberately chosen simulation parameters. They should not be interpreted as universal decision boundaries.

Third, the Jaccard threshold is not selected using an optimization criterion. The threshold sensitivity analysis is intended to characterize the operating trade-off rather than identify a preferred value.

Fourth, the simulation uses the final interim analysis as a reference point for measuring reversal. This is useful for studying sequential instability, but it is not equivalent to evaluation against an independent external sample or the true treatment-effect function.

Fifth, the stopping rule is based on point estimates and does not incorporate formal sequential Type-I error control, confidence sequences, alpha spending, or multiplicity adjustment. Consequently, the results should not be interpreted as demonstrating valid repeated-look inference.

Finally, the simulation contains only 500 replicated experiments. The reported percentages are therefore Monte Carlo estimates and would themselves have sampling uncertainty. Increasing the number of replications would improve precision, particularly for operating characteristics near zero or one.

9 Conclusion

This study examined whether the stability of an estimated harmful population can provide useful information for sequential A/B decisions under treatment effect heterogeneity.

The proposed stability-augmented rule combines a minimum harmful fraction with a minimum Jaccard similarity between consecutive estimated harmful populations. In the threshold scenario, where the estimated harmful population is comparatively unstable, the additional requirement reduces decision reversal and increases agreement with the final interim analysis, while increasing the mean stopping sample size. In the smooth scenario, where the estimated harmful population is already persistent, the stability requirement has much less effect.

The central finding is therefore qualitative rather than threshold-specific: *stability of the estimated affected population is an informative dimension of sequential heterogeneous-effect decisions.* It captures a property that is not represented by the harmful fraction alone.

The present study does not establish formal inferential validity or an optimal stability threshold. A natural next step is to integrate the stability criterion with formally calibrated sequential inference, independent evaluation of discovered subgroups, and more flexible CATE estimators. Such extensions would determine whether the descriptive stability signal observed here can be converted into a statistically controlled decision procedure.

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A Reproducibility

The complete simulation workflow is implemented in R and is organized into scripts for data generation, CATE estimation, sequential analysis, repeated simulation, baseline stopping rules, stability-based stopping, stability metrics, threshold sensitivity analysis, and final figure generation.

The simulation uses a fixed seed for the controlled data-generation and validation stages. The repeated simulation framework generates independent replicates and evaluates the same analysis procedure at the six cumulative interim sample sizes.

The repository contains the generated CSV results and figures used in this manuscript. The compiled PDF is provided separately from the LaTeX source in the public repository.